

Model Explainer: Equipment Failure Prediction

Week 9 — Operational ML Deliverable | Prepared for the Maintenance & Operations Team | [Your Name]

The Analogy

Think of the model as an experienced maintenance technician who has quietly watched every machine on the floor for years and kept a mental notebook of what conditions tend to come right before a breakdown. Every morning, that technician walks the floor, glances at each machine's temperature, vibration, pressure, wear, and recent alarm history, and posts a sticky note on the machines that “feel off” based on everything they've seen before — not a certainty, just an experienced hunch worth checking. The model does the same thing, at scale, for every machine, every day: it doesn't know the future, it recognizes patterns that have historically preceded failure and raises a flag. Like that technician's hunch, the flag is a prompt to go look closer, not a verdict.

The Drivers

The chart below shows which sensor readings the model leans on most heavily when deciding a machine's risk level, averaged across the full test fleet.

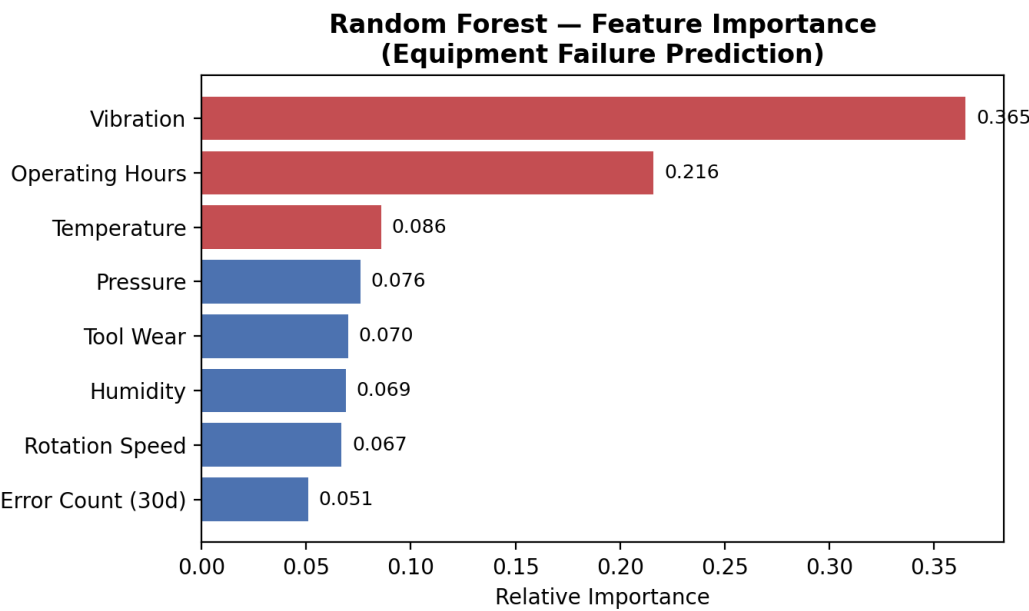


Figure 1. Random Forest feature importance for the equipment-failure model. Red bars mark the top three drivers.

Top 3 drivers, in plain English:

- 1. Vibration:** By far the strongest signal. A machine that is starting to vibrate more than its normal baseline is very often showing early mechanical wear — a loosening part, a bearing starting to go, or a misalignment. This is the same thing an experienced technician listens and feels for on the floor, which is why the model leans on it so heavily.
- 2. Operating hours (cumulative run time):** The longer a machine has been running since its last major service, the more its risk climbs. This lines up with basic wear-and-tear intuition: an older, harder-worked

machine is simply more likely to be due for a failure than a freshly serviced one.

3. Temperature: Running hotter than the normal operating range is often a downstream symptom of friction, restricted airflow, or a component working harder than it should — so temperature acts as a secondary confirming signal alongside vibration.

Trust & Limits

What the model gets wrong, and why that's expected: No model is perfect, and given how rare failures are relative to normal operation, this one will make two kinds of mistakes:

Error type	What it means on the floor	Practical impact
False positive (false alarm)	Model flags a machine as high-risk, but it was actually fine.	A technician spends time inspecting a healthy machine — low cost, minor inconvenience.
False negative (missed failure)	Model does not flag a machine that goes on to fail.	Unplanned downtime, possible safety issue, costlier emergency repair — the error we most want to avoid.

Because a missed failure is far more expensive than a false alarm, the model was deliberately tuned to favor catching more true failures, even if that means more false alarms than a default, accuracy-optimized model would produce. That tradeoff was a conscious choice, not an oversight.

How operations should use this tool:

- Treat the model's output as a **prioritized watchlist**, not an automatic shutdown or repair order. A “high risk” flag means “inspect this one next,” not “this machine has failed.”
- Always pair the flag with the technician's own judgment and physical inspection. The model has never touched the machine — it only sees eight sensor numbers.
- Expect and budget for some false alarms; they are the cost of catching more real failures early.
- Revisit and retrain the model periodically. Equipment ages, maintenance practices change, and sensors drift — a model trained on last year's patterns will slowly become less accurate if left untouched.
- If a machine is flagged repeatedly with no fault found, log it — that pattern itself is useful feedback for improving the model over time.

In short: this model is a force multiplier for an experienced maintenance team's attention, not a replacement for their judgment.